



Review

The current landscape of learning analytics in higher education

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ABSTRACT

Learning analytics can improve learning practice by transforming the ways we support learning processes. This study is based on the analysis of 252 papers on learning analytics in higher education published between 2012 and 2018. The main research question is: What is the current scientific knowledge about the application of learning analytics in higher education? The focus is on research approaches, methods and the evidence for learning analytics. The evidence was examined in relation to four earlier validated propositions: whether learning analytics i) improve learning outcomes, ii) support learning and teaching, iii) are deployed widely, and iv) are used ethically. The results demonstrate that overall there is little evidence that shows improvements in students' learning outcomes (9%) as well as learning support and teaching (35%). Similarly, little evidence was found for the third (6%) and the fourth (18%) proposition. Despite the fact that the identified potential for improving learner practice is high, we cannot currently see much transfer of the suggested potential into higher educational practice over the years. However, the analysis of the existing evidence for learning analytics indicates that there is a shift towards a deeper understanding of students' learning experiences for the last years.

1. Introduction

The pervasive integration of digital technology into higher education (HE) influences both teaching and learning practices, and allows access to data, mainly available from online learning environments, that can be used to improve students' learning. Online learning facilitating the use of asynchronous and synchronous interaction and communication within a virtual environment (Broadbent & Poon, 2015), has “succeeded in becoming an integral part of HE, and now it needs to turn its focus, from providing access to university education, to increasing its quality” (Lee, 2017, p. 15). To this end, HE institutions are implementing Learning Analytics (LA) systems to better understand and support student learning (Schumacher & Ifenthaler, 2018). This study presents a literature review with the objective of mapping the current landscape of contemporary LA research in HE.

There is an evolving interest in LA among not only practitioners but also researchers in Technology-Enhanced Learning (TEL). LA emerges as a fast-growing and multi-disciplinary area of TEL (Ferguson, 2012), which forms its own domain (Strang, 2016). In LA, information about learners and learning environments is used to “access, elicit, and analyse them for modelling, prediction, and optimization of learning processes” (Mah, 2016, p. 288).

Definitions of LA vary. Some define it explicitly in terms of the use

of student-generated data for the prediction of educational outcomes, with the purpose of tailoring education (Junco & Clem, 2015; Xing, Guo, Petakovic, & Goggins, 2015). Others define LA as a means to help educators examine, understand, and support students' study behaviours and change their learning environments (Drachler & Kalz, 2012; Rubel & Jones, 2016). While there is no generally accepted definition of LA, many refer to it as “the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs” (Long & Siemens, 2011, p. 34).

LA, academic analytics, and educational data mining (EDM) are closely related research areas. The goal of academic analytics is to support the institutional, operational, and financial decision-making processes (Lawson, Beer, Rossi, Moore, & Fleming, 2016), while the overall purpose of LA and EDM is to understand how students learn. Based on the analysis of large-scale educational data, LA and EDM aim to support research and practice in education (Siemens & Baker, 2012). Both EDM and LA reflect the emergence of data-intensive approaches to education and there are similarities between EDM and LA, which suggests several areas of overlap (Siemens & Baker, 2012). There are also, however, several distinctions between them (Siemens & Baker, 2012). First, one key distinction concerns the type of discovery that is prioritised: EDM has a primarily focus on automated discovery, whereas LA

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has a stronger focus on leveraging human judgement. Second, EDM models are often used as the basis for automated adaptation, conducted by a computer system, whereas LA models are often developed to inform instructors and learners. Third, EDM researchers use reductionist frameworks: they reduce phenomena to components and focus on the analysis of individual components and relationships between them. By contrast, LA researchers have a stronger focus on understanding complex systems as wholes. In this study, we are particularly interested in how LA research has been employed across different higher educational settings, disciplines, institutional types and states.

1.1. Research background

Although the LA research field is still in its infancy, it has already been the focus of a number of literature reviews, which are helpful, but are mainly aimed at researchers and not practitioners (Ferguson & Clow, 2017). Some of the reviews focus explicitly on the use of LA in higher educational settings (e.g., Avella, Kebritchi, Nunn, & Kanai, 2016; Clow, 2013; Ferguson & Clow, 2017; Ithantola et al., 2016; Leitner, Khalil, & Ebner, 2017; Sin & Muthu, 2015), whereas others focus on educational contexts in general (e.g., Dawson, Gašević, Siemens, & Joksimovic, 2014; Ferguson, 2012; Ferguson et al., 2016; Jivet, Scheffel, Specht, & Drachsler, 2018; Nistor, Derntl, & Klamma, 2015; Papamitsiou & Economides, 2014; Peña-Ayala, 2018).

The reviews focusing explicitly on HE have already identified several important aspects of the LA research to be considered. Avella et al. (2016), for example, examined LA methods, benefits for and challenges in HE. Some of the common methods used are data visualisation, social network analysis, prediction and relationship mining. Whilst data tracking, collection and evaluation were highlighted as some of the challenges associated with LA research, targeted student learning outcomes and behaviour were mentioned as potential benefits. Another recent field overview (Leitner, Khalil, & Ebner, 2017) focused on the analysis of the current research trends of LA, limitations, methods and the key stakeholders. The results showed that the usage of massive online open courses (MOOCs), enhancement of learning performance, student behaviour, and benchmarking of learning environments were the key areas of the LA' research focus; the limitations included the needed time to prepare data or getting the results, the size of the available dataset and examined group, and ethical reasons. Among the used methods, prediction, distillation of data for human judgement, and outlier detection were found to be the most common methods used in the HE domain. The main identified stakeholders are researchers, rather than learners. This contradicts with the adopted in our study and by a large number of LA scholars definition of LA (Long & Siemens, 2011) that emphasises mainly learners and their learning environments.

One of the latest reviews (Ferguson & Clow, 2017) included the exploration of the evidence of whether LA improve learning practice in HE based on four propositions of LA: 1) they improve learning outcomes, 2) they support learning and teaching, 3) are deployed widely, and 4) are used ethically. Based on these propositions, the authors pinpointed that many studies did not contain strong evidence *for* or *against* one of these propositions. Most of the evidence based on the analysis of 28 papers relates to the proposition that LA improve learning support and teaching, comprising retention, completion and progression, which has been categorised as evidence that LA improve teaching in universities. The weaknesses of the present research include lack of geographical spread, gaps in knowledge (e.g., in terms informal learning and a lack of negative evidence), little evaluation of commercially available tools, and little attention to ethics. Some other studies (Ithantola et al., 2016; Sin & Muthu, 2015) examined both LA and EDM research in HE confirming several of the findings presented above.

Whereas the above-discussed research was conducted with a focus on HE explicitly, the studies discussed below target the educational context broadly, including the review of the studies conducted in the

HE context. One of the earlier reviews (Ferguson, 2012) investigates the technological, educational and political factors driving the development of LA in education. This review elaborates on the discussion about the relationships between LA, EDM, and academic analytics. Dawson, Gašević, Siemens, and Joksimovic (2014) performed a citation network analysis to identify the emergence of trends and disciplinary hierarchies that influence the field's development. The results show that the most commonly cited papers are conceptual and review-based, which implies a need for scholars to define a space for LA. This is also confirmed by Ferguson and Clow (2017). Papamitsiou and Economides (2014) conducted a systematic literature review of empirical LA and EDM research. Most of the studies were found to be explorative or experimental and conducted within virtual learning environments (VLE) or LMSs. Drachsler and Kalz (2016) present a reflective summary of the ongoing research on LA and MOOCs and provide a conceptual framework to position current research on MOOCs and the LA innovation cycle. One of the recent reviews (Jivet et al., 2018) investigated the extent to which theories and models from learning sciences have been integrated into the development of learning dashboards aimed at learners. The results show that: i) dashboards evaluations rarely consider concepts from learning sciences, ii) there is a low number of validated instruments used to assess either the learners' skills or the tools, and iii) the major focus is on evaluating a dashboard's acceptance, usefulness and ease of use as perceived by learners, rather than on whether the dashboard brings any benefit to learners. Finally, Peña-Ayala (2018) conducted a detailed review of LA with the aim of providing an idea of the LA toil, its research lines, and trends to inspire the development of novel approaches for improving teaching and learning practices. This study shows that a small number of research studies proposed to acknowledge the progress of the field provides representative concepts and determines the role of the diverse stakeholders. Furthermore, it pinpoints the need of a well-sounded theoretical frame to guide the LA endeavour.

Even though there have been attempts to analyse various aspects of LA research both in HE and education in general, none of these studies have systematically structured and summarised the publications related to LA in HE on a larger scale with a specific focus on the examination of research methods, approaches and the evidence of whether LA improve learning practice in HE. We aim to fill in this gap by presenting an extensive coverage of research that concerns the use of LA in HE and that aims to target not only researchers but also practitioners.

1.2. Research questions

The main research question is:

What is the current scientific knowledge about the application of learning analytics in higher education?

This overall research question is operationalised by the following sub-questions:

- What research approaches and methods are used?
- What is the evidence of the learning analytics research?

2. Method

We followed Webster and Watson's (2002) method for literature reviews covering publications from 2012 to 2017 and also included the LAK conference proceedings from 2018. We chose these years since LA is a young field and its adoption and the emergence of peer-reviewed journal articles and conference proceedings has increased since 2012.

2.1. Literature search strategy

We initially searched for relevant publications through PRIMO (Peer-Reviewed Instructional Materials Online Database), which includes numerous databases, such as Web of Science and Scopus. To

ensure reliability we followed Webster and Watson's (2002) guidelines which suggest starting with contributions published in leading journals, when identifying relevant literature. Consequently, we manually searched for LA papers in four key high-ranked journals, namely *Computers in Human behaviour*, *the Internet and Higher Education*, *Computers & Education*, and *British Journal of Educational Technology*. To further complement the study's data set we also manually checked four special issues on LA, published by the *Journal of Asynchronous Learning Networks* (2012, vol. 16, issue 3), the *Journal of Educational Technology and Society* (2012, vol. 15, issue 3), the *American Behavioural Scientist Journal* (2013, vol. 57, issue 10), the *Online Learning Journal* (2016, vol. 20, issue 2), *Technology, Knowledge and Learning* (2017, vol. 22, issue 3). Articles published in the field-specific *Journal of Learning Analytics* were also included. In order to offer a comprehensive picture of LA research, we also included the proceedings of the International Learning Analytics and Knowledge (LAK) conferences for the years 2012–2018. This conference was chosen as it serves as a forum for LA research and provides a solid ground for analysing emerging LA research.

In the databases we searched for “learning analytics” and “higher education” in the articles’ titles, keywords and/or abstracts. To ensure reliability and validity we carefully examined the title, abstract and keywords of the articles. The initial search resulted in 1434 papers, but after applying our selection criteria, listed below, the final data set comprised 252 research papers.

The selection of papers was based on the following criteria:

1. We included empirical and theoretical research studies that focus on LA in HE.
2. We excluded LA review articles that explicitly aimed to cover LA as a research field in a systematic way.
3. We excluded papers with a focus on LA and MOOCs, as the context is so different (e.g., time pressure, social context, teacher-student relations) from campus environments. Also, there is already fairly recent summaries of ongoing research (see e.g., Drachsler & Kalz, 2016; Leitner et al., 2017).
4. Studies with a single emphasis on academic analytics or EDM were omitted. However, we have included papers where the fields of LA, academic analytics and EDM explicitly overlap, represented either in the articles' titles, abstracts or keywords.
5. For the conference proceedings, we only included papers published as part of the main conference. Workshop papers and posters were excluded.
6. Only peer-reviewed papers published in English during the period January 2012–March 2018 were included.

2.2. Data analysis

All studies were analysed in order to assess the research in terms of the studies’ research approaches, methods of data collection and analysis, and the evidence of whether LA improve learning practice in HE. The papers were coded independently by all the authors. Five percent of all reviewed articles were coded by the authors together to verify the coding. When discrepancies in the coding were found we discussed the differences and re-coded the papers until we agreed on the mapping.

2.2.1. Research approaches

The analysis of research approaches followed a categorisation developed by Grönlund and Andersson (2006) and later used in other studies (e.g., Viberg & Grönlund, 2013). The categories for the research approaches are presented in Table 1.

2.2.2. Methods of data collection

The analysis of methods of data collection similarly followed the categorisation presented by Grönlund and Andersson (2006). When examining methods of data collection, we categorised them according to single versus mixed method of collections as papers could belong to

Table 1
Research approaches. Categories adopted from Grönlund and Andersson (2006).

Research approach	Definition
Descriptive	Describes a phenomenon in its appearance without any use of theory
Philosophical	Reflects upon a phenomenon without data and any use of theory
Theoretical	Reflects on a phenomenon based on some theory but without empirical data
Theory use	Applies a theory/theories or models as a framework for the conducted study
Theory generating	Analyse data in a systematic manner with a purpose of building a theory
Theory testing	Test a theory using data in systematic manner

several categories. For example, a paper could be both interpretative and use a survey, or both a product description (i.e., a new LA tool) and an experiment. Table 2 presents the categories we used for the methods of data collection.

2.2.3. Methods of data analysis

When examining methods of data analysis, we first categorised them according to single versus mixed method of analysis (i.e., qualitative and quantitative methods are used in the same study). Secondly, as the majority of LA research applies computational methods, we focused our analysis on the examination of these methods, and followed the categorisation suggested by Merceron (2015) (See Table 3).

2.2.4. Research evidence in the field of LA

Examining the evidence of whether LA improve learning practice in HE, we adopted the four validated propositions by Ferguson and Clow (2017) mentioned in section 1.1 to structure evidence in the field of LA. When determining whether papers fulfilled Ferguson and Clow (2017) four propositions, we used three classes: “yes”, “no”, and “potentially”, where the last should be interpreted as not being able to identify evidence supporting this proposition, but the authors made arguments in the discussion that their results could lead to improvements in the future.

3. Results

In this literature review, 252 papers are included (Fig. 1): 136 papers (54%) are conference papers and 116 journal publications (46%).

3.1. Research approaches

Our analysis shows in Fig. 2 that most papers (57%) in our sample undertake a *descriptive research* approach. For example, Lonn, Krumm, Waddington, and Teasley (2012) illustrate an early warning system (EWS) for a mentoring program that supports students. Santos, Verbert, Govaerts, and Duval (2013) describe an evaluation study of the use of a dashboard and the extent to which it addresses the students’ needs. The descriptive approach is frequently associated with studies that present or evaluate a specific LA tool (e.g., Dimopoulos, Petropoulou, & Retails, 2013), with experimental studies (e.g., Ali, Hatala, Gašević, & Jovanovic, 2012), as well as interpretative studies (e.g., Nguyen, Rienties, Toetenel, Ferguson, & Whitelock, 2017; Waddington, Nam, Lonn, & Teasley, 2016).

Another frequent approach is *theory use research* (26%). In our review, we use a broad definition of theory and it “encompasses theoretical frameworks, models, theories (as a body of knowledge in a broader sense) and theoretical concepts used” (Viberg, 2015, p. 43). We found 62 different theories used. There are no dominating theories; rather there is a plethora of theories used to explain different aspects of LA, such as human behaviour (Gilmore, 2014), learning and knowledge

Table 2
Methods of data collection. Categories adopted from Grönlund and Andersson (2006).

Methods	Description
Argument	Logical argument but not based in any particular theory or relating explicitly or by clear implication to any theory.
Ethnography	Any attempt to understand actions by systematic observation and interpretation.
Experiment	Field and quasi- experiments included. This category applies to systematic/structured tests even though in field settings many environmental variables clearly are not controlled, and tests may be rather explorative.
Interpretative	Any kind of data collection more strictly performed than a “case story” but not necessarily a strictly explained or described method for interpretation. Case study belongs here, as do more limited studies where qualitative and quantitative data is used. Studies that use data from large data sets, e.g., LMS, for the purpose of interpreting the studied phenomenon.
Literature study	Only documents used. Not necessarily a strict method or even explicitly labelled as a literature study.
Product description	IT product, method or similar, described by the manufacturer or someone else.
Survey	This includes qualitative overviews of several documents and cases as well.
Unclear	Even the widely defined categories above fail to capture the method.

outcomes (Kovanović et al., 2015), or technology acceptance and use (Nistor et al., 2014). While the theoretical development of the field is still in its infancy, a few field-specific theories have been developed and applied. Agudo-Peregrina, Iglesias-Pradas, Conde-González, and Hernández-García (2014), for example, use the conceptual framework of analytics in education, developed by Van Barneveld, Arnold, and Campbell (2012). Tempelaar, Rienties, and Giesbers (2015) apply Buckingham Shum and Crick's (2012) framework of dispositional LA to investigate the predictive power of learning dispositions, outcomes of formative assessments and other system-generated data in modelling students' performance.

Twenty-five papers (11%) show evidence of *theory generation* (Fig. 2). These papers analyse data with the purpose of taking steps forward in theory building. de Freitas, Gibson, Du Plessis, Halloran, Williams, Ambrose et al. (2015) propose a foundational LA model of HE, focusing on the dynamic interaction of stakeholders with data supported by visual analytics. West, Heath, and Huijser (2016) present the *Let's Talk Learning Analytics Framework* of factors relevant to the institutional implementation of LA. Wise, Zhao, and Hausknecht (2013) explicate a pedagogical model for analytics interventions based on the principles of integration, diversity, agency, reflection, parity, and dialogue. Lang, Macfadyen, Slade, Prinsloo, and Sclater (2018) presented “LA Code of Ethics v1.0” and sought input and feedback on this draft code from individual practitioners across the LA spectrum.

Four percent of papers are *philosophical* (Fig. 2), reflecting on LA without any data or theory use. Daniel (2015), for instance, identifies current challenges facing HE institutions and explores the Big Data potential in addressing these challenges. The potential of LA to support a shift in focus from the assessment of individual student performance in isolation to the assessment of their performance as a team are discussed by Williams (2017).

Two percent of studies are *theoretical* (see Picciano, 2012; Prinsloo & Slade, 2016; Arnold & Sclater, 2017; Lodge, Alhadad, Lewis, & Gašević, 2017; Prinsloo & Slade, 2017). Only one study attempts to *test a theory* (Ali, Asadi, Gašević, Jovanovic, & Hatala, 2013).

3.2. Methods of data collection

In the papers analysed, methods of data collection such as interpretative study, experiment, literature study, survey, product description, argument and ethnography are used (see Fig. 3; the total percent is higher than 100% since many papers use more than one method).

The majority (72%) of the papers use a single method of data collection and only 28% of the papers employ mixed data collection methods (see Fig. 4).

Irrespective of the studies use a single or mixed method of data collection, the most common method is interpretative methods, 68% of the studies use interpretative data collection methods, such as interviews or focus groups (e.g., McCoy & Shih, 2016; Tsai, Moreno-Marcos, Tammets, Kollom, & Gašević, 2018). We have broadened the definition of the interpretative studies, presented by Grönlund and Andersson (2006), by including studies that used data from large data sets – for example, from LMS – for the purpose of interpreting the studied phenomenon (e.g., Rienties & Toetenel, 2016). The second most commonly applied method is experiment (18%), the third one is product description (15%), followed by surveys (11%). The studies that used mixed methods for data collection mainly concerned a specific LA tool that was introduced and further evaluated through some interpretative method, such as a case study. Thus, those papers were classified both under the ‘product description’ category and also as ‘interpretative’ studies. Additionally, some authors presented a tool and evaluated it through an experiment (e.g., Ali et al., 2012). Surveys and interpretive methods of data collection are also among the commonly used combinations for data collection (e.g., Rodríguez-Triana, Prieto, Martínez-Monés, Asensio-Pérez, & Dimitriadis, 2018; Santos et al., 2013). In all, 72% of the papers combining different methods of data collection mixed interpretative studies with some other method.

3.3. Methods of data analysis

A mixed method approach, i.e., the studies where both qualitative and quantitative methods of analysis were used, was employed in 19% of all papers. The application of the mixed methods for data analysis

Table 3
Computational methods for data analysis. Categories adopted from Merceron (2015).

Methods	Description
Prediction	A major task tackled by prediction methods is to predict performance of students. The most common methods include regression and classification.
Clustering	Clustering techniques are used to group objects so that similar objects are in the same cluster and dissimilar objects in different clusters.
Relationship Mining	This category includes such methods as association rule mining, correlation mining, sequential pattern mining and casual data mining.
Distillation of data for human judgement	This category includes statistics and visualizations that help humans make sense of their findings and analyses.
Discovery with models	This category encompasses approaches in which the model obtained in a previous study is included in the data to discover more patterns.

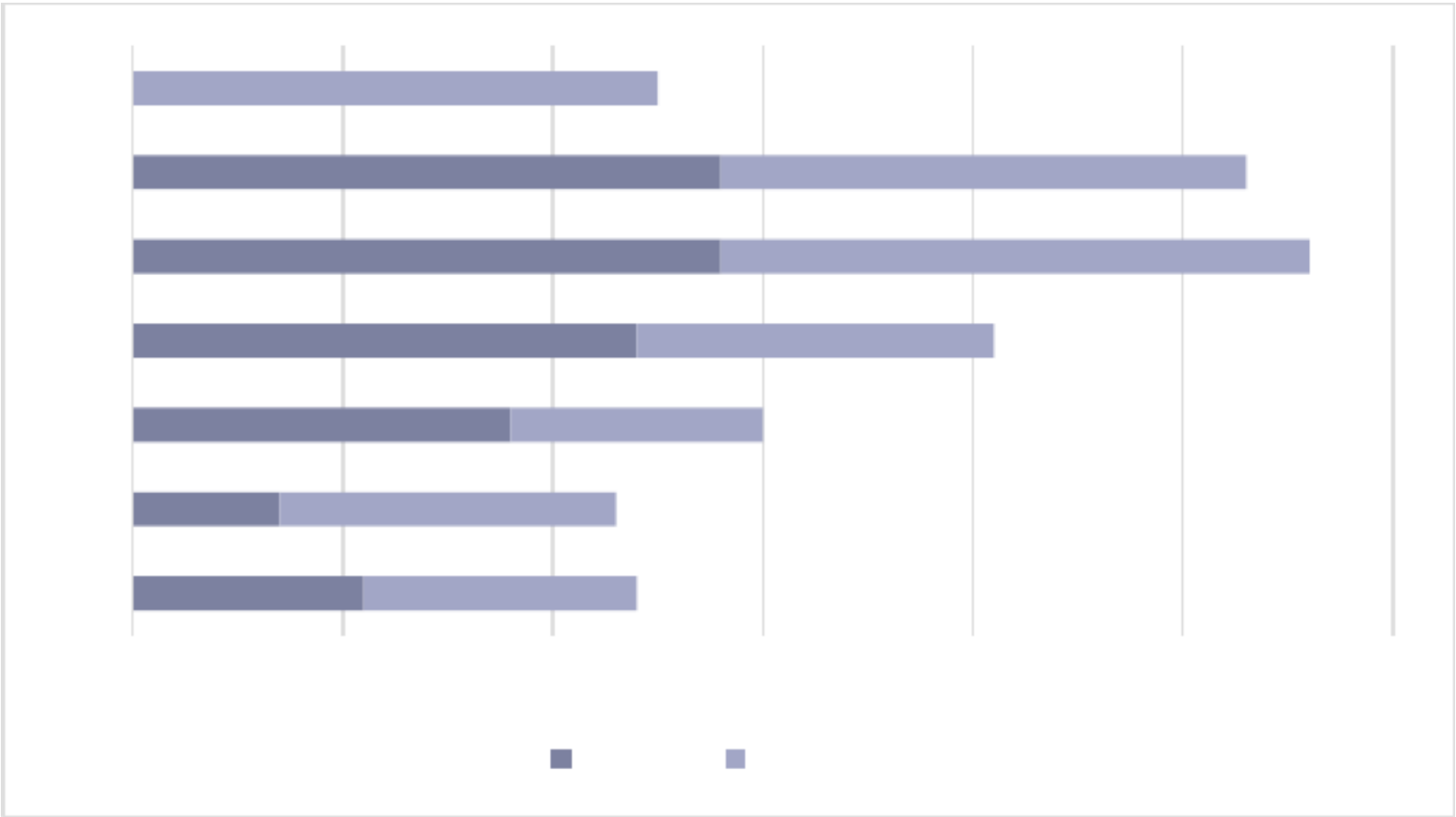


Fig. 1. Number of papers included in the literature review.

has increased to 40% 2017, compared to 18% 2016, 7% 2015 and 13% 2014. The 2018 LAK conference has 36% of the studies applying both qualitative and quantitative methods.

Following Merceron's categorisation (Table 3), our results show that predictive methods (including regression and classification) were the most frequent methods (32%), followed by relationship mining (24%) and distillation of data for human judgement (24%) including statistics and visualisations that help people make sense of their findings. Finally, 37 studies (15%) applied the discovery-with-models approaches and 28 papers (11%) used clustering techniques. As shown in Fig. 5, predictive methods of data analysis were one of the most common methods in 2012 (25%) and 2013 (29%) and leading in 2014 (47%), 2015 (56%), but the use of such methods has significantly decreased for the years 2016 (20%) and 2017 (15%). The higher number (48%) for 2018 does not illustrate a whole picture, as the sample for this year is limited. Moreover, there is an increase for the relationship mining methods for the year 2017 (51%) compared to 2016 (14%), 2015 (24%) and 2014 (8%). The relative popularity between the methods is stable over the years.

3.4. What is the evidence of the LA research?

In this section, the findings of the LA research in terms of its evidence for learning and teaching in HE are presented. Overall, the results of our analysis show that there is little evidence in a stronger sense, i.e., when the reviewed papers' results show improvements – in, for

example, students' learning outcomes as well as learning support and teaching - about LA in higher educational settings (Fig. 6). This evidence is presented by the “yes” category in Fig. 6. The “potentially” category includes studies that were categorised as suggesting, in many papers explicitly, that there is the potential to improve e.g., learning outcomes and/or to support teaching. The presentation of the results below focuses on the evidence of LA in the stronger sense. The findings are presented according to the four propositions (Ferguson & Clow, 2017) mentioned in section 1.1.

- 1. LA improve learning outcomes,
- 2. LA improve learning support and teaching,
- 3. LA are taken up and used widely, including deployment at scale,
- 4. LA are used in an ethical way.

The proposition with most evidence (35%) in LA is that *LA improve learning support and teaching in higher education* (Fig. 6). There is little evidence in terms of improving students' learning outcomes. Only 9% (23 papers out of all the 252 reviewed studies) present evidence in this respect. Moreover, there is even less evidence for the third proposition. In only 6% of the papers *LA are taken up and used widely*. This suggests that LA research has so far been rather uncertain to this proposition. Finally, our results reveal that 18% of the research studies even mention “ethics” or “privacy” (Fig. 6). This is a rather small number considering that LA research, at least its empirical strand, should seriously approach the relevant ethics.

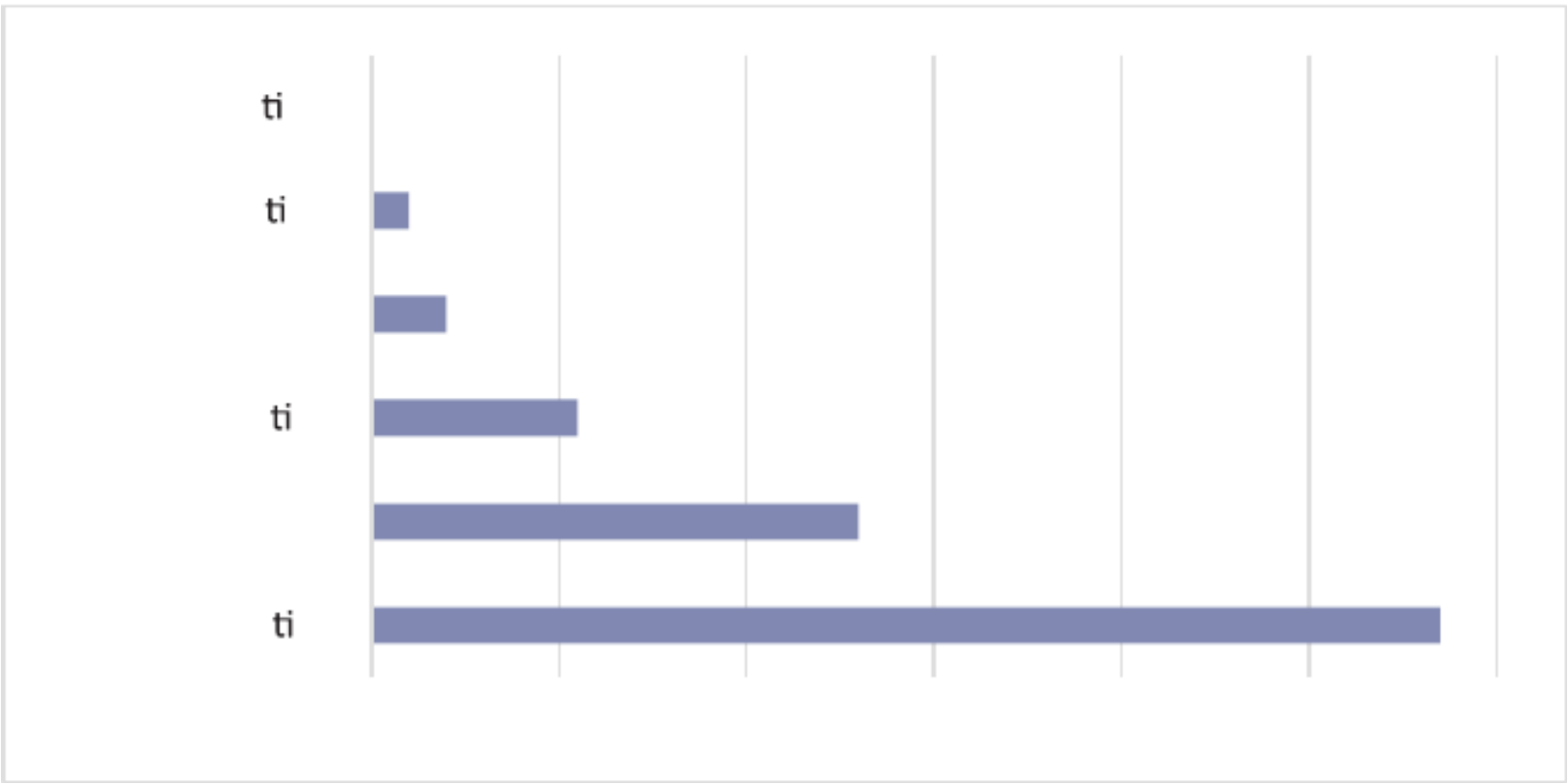


Fig. 2. Research approaches (%).

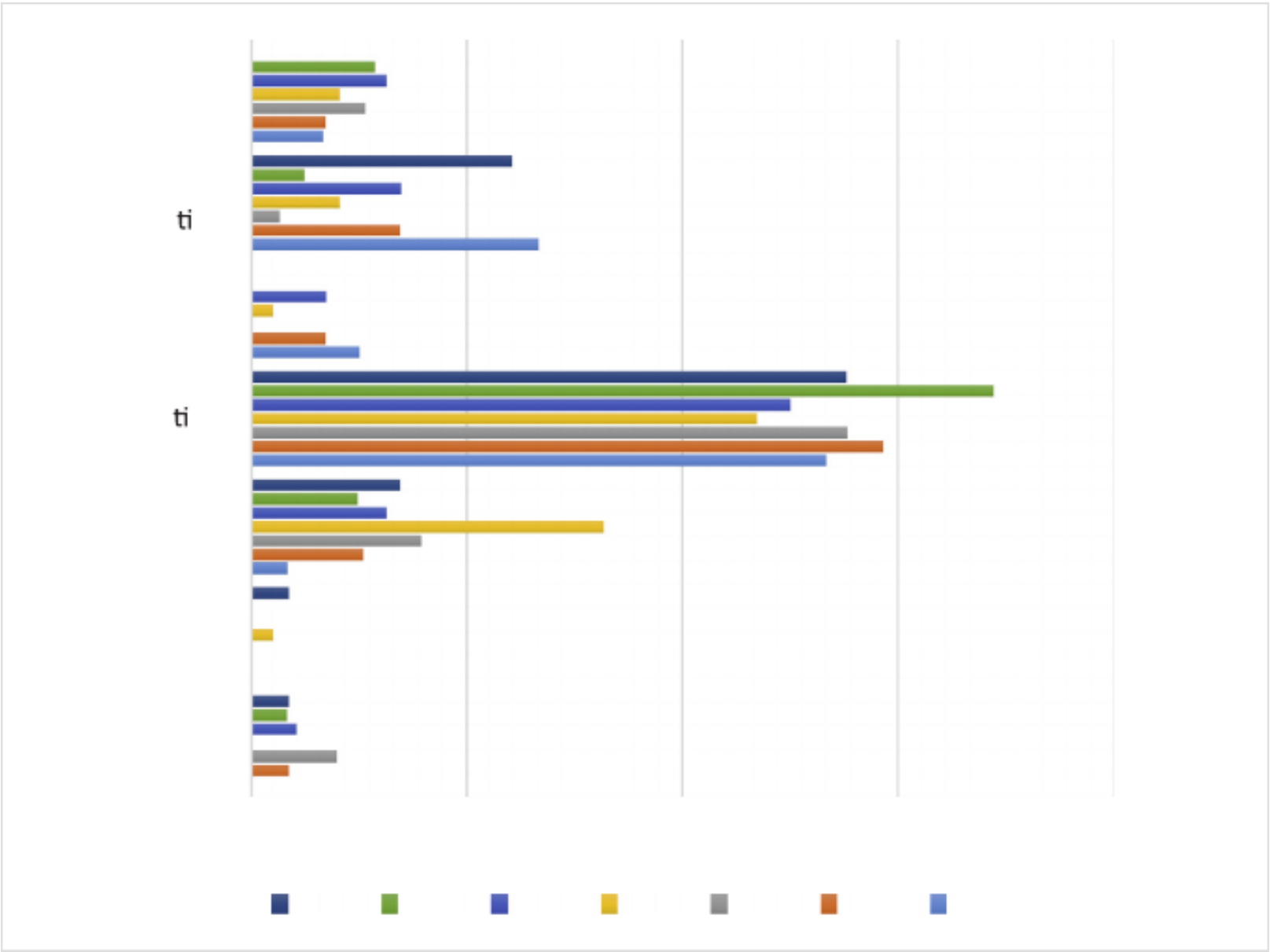


Fig. 3. Methods of data collection.

Our results also demonstrate that the overall potential of LA is so far higher than the actual evidence (Fig. 6) for the first two propositions. The potential of LA to improve learning outcomes is significantly higher than its current evidence and the potential of LA to improve learning support and teaching is even higher, compared to what has been actually exhibited (Fig. 6). For the last two propositions we examined the research papers only with a focus on its evidence.

Furthermore, the results show that the evidence in its stronger sense for LA across the years 2012–2018 has been rather equally distributed (Fig. 7). The evidence for the proposition that *LA improve learning support and teaching*, for example, has been dominating across all the years: 2012 (33%), 2013 (30%), 2014 (33%), 2015 (39%), 2016 (41%), 2017 (34%), 2018 (20%). The numbers for the last year are not final yet, as our findings for this period are only based on the analysis of the LAK conference papers published in March 2018. The more detailed results for each proposition are presented in the following sections.

Overall, we found only one study (Kwong, Wong, & Yue, 2017) that

falls under all four propositions. There is a small number of papers that support the propositions 1 and 2, as for example, the case of Nguyen, Huptych, and Rienties (2018) who focus on the intersection of learning design and academic performance, the case of Worsley (2018) that focused on student engagement and the identification of efficacious learning practices, and the study of Tempelaar, Rienties, and Nguyen (2017) who examined learning strategies in conjunction with the use of worked examples. There are even less studies that support three propositions (e.g., Broos, Verbert, Langie, Van Soom, & De Laet, 2018; Hart, Daucourt, & Ganley, 2017; Millecamp, Gutiérrez, Charleer, Verbert, & De Laet, 2018).

3.4.1. Do LA improve learning outcomes?

One of the larger expectations of LA research and practice relates to the proposition that it will improve students' learning outcomes. However, this proposition has been confirmed by only few studies (9%) (Fig. 6). In our sample, we found only one study from 2012 (Arnold &

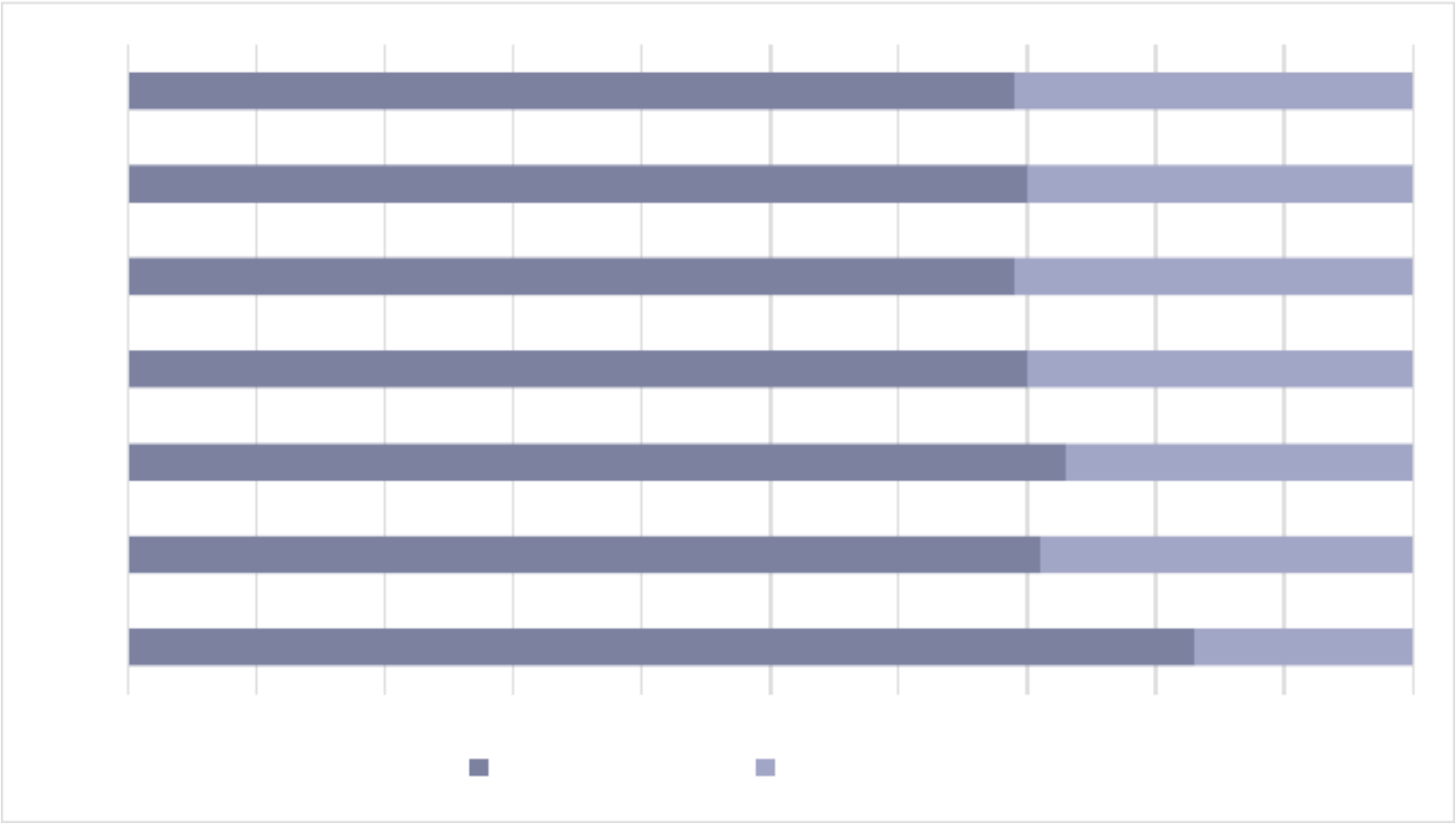


Fig. 4. Single-vs Mixed methods of data collection in percentage per year.

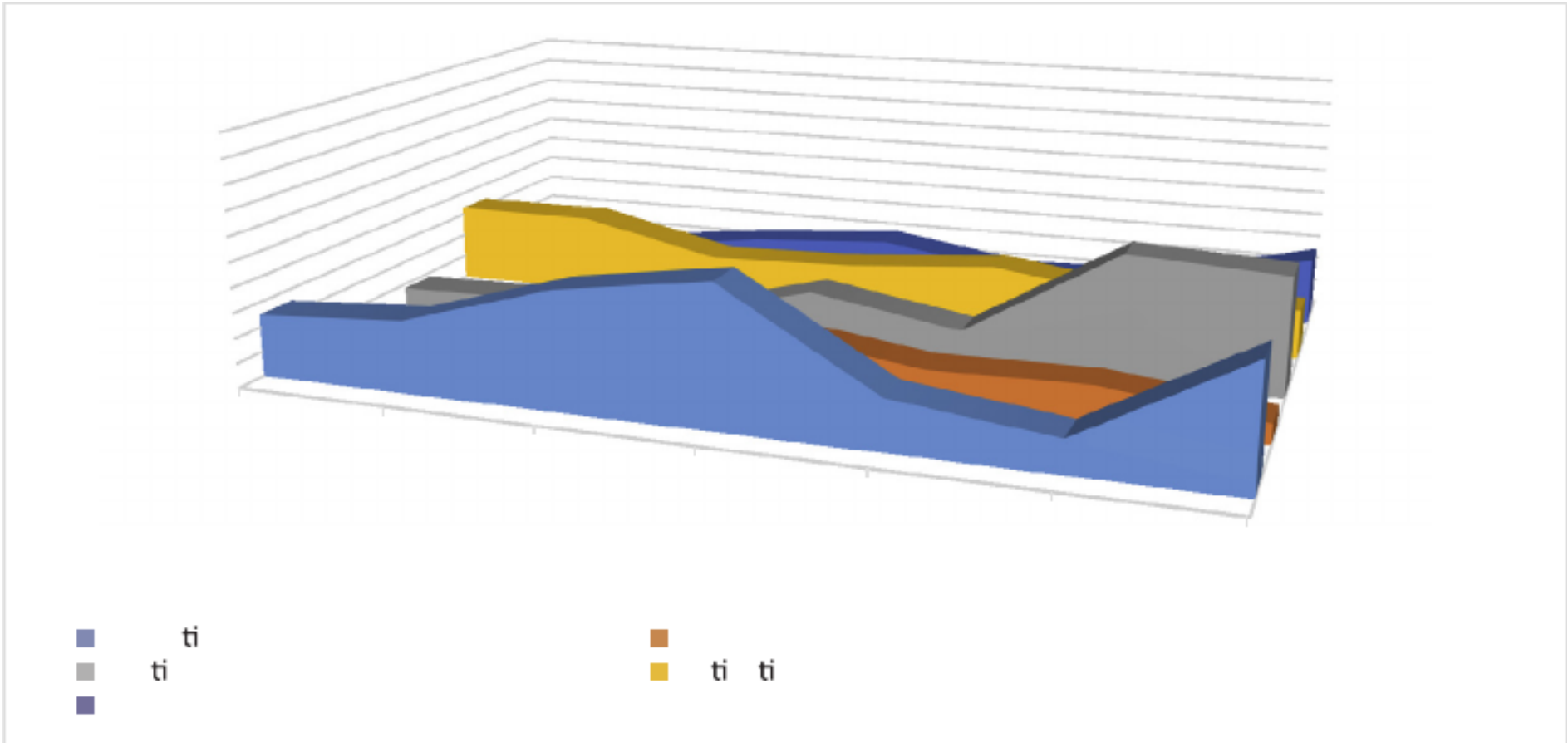


Fig. 5. Computational methods for data analysis (%).

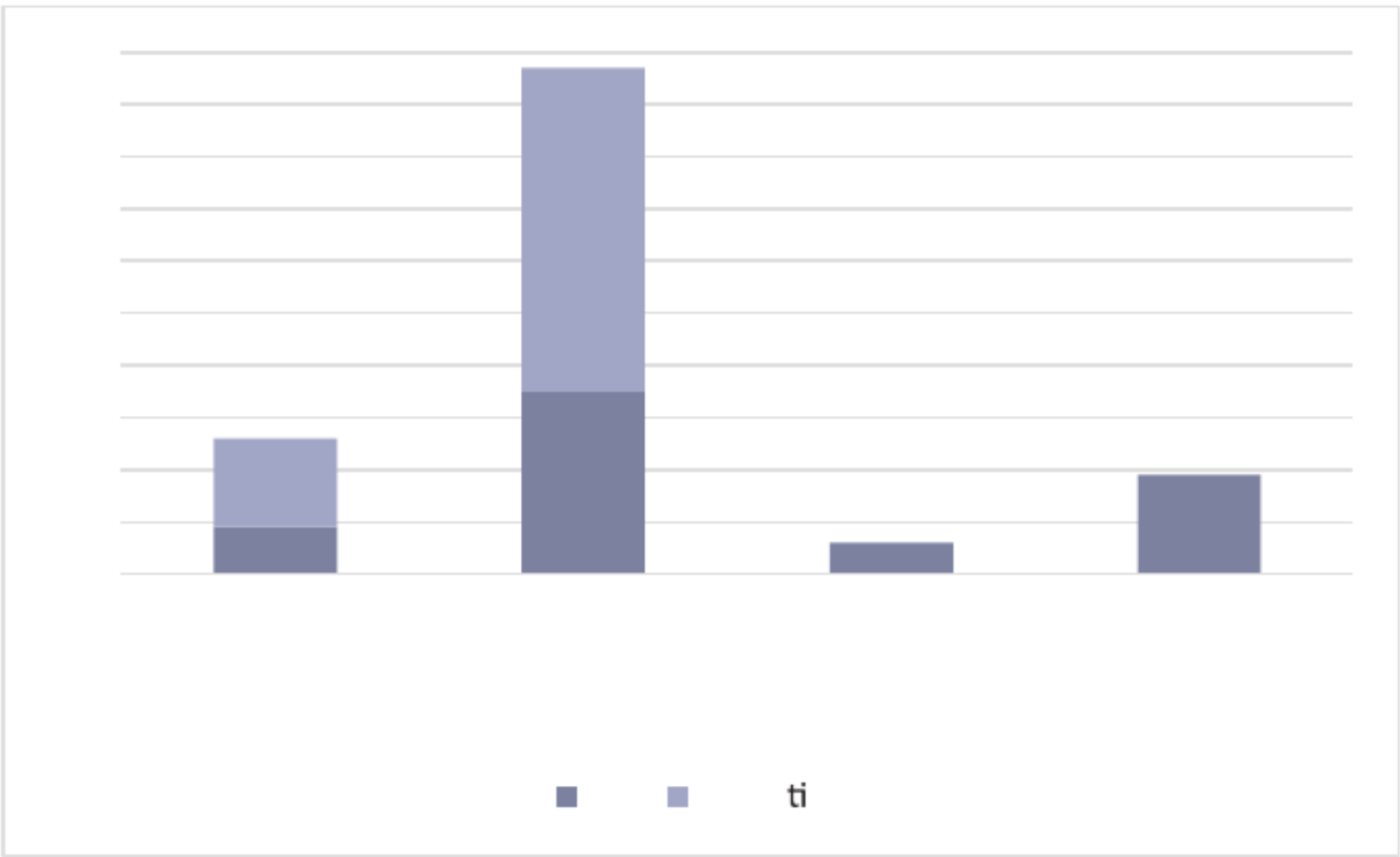


Fig. 6. Evidence for learning analytics in higher education (%).

Pistilli, 2012) that strongly supports this proposition. The papers published from 2013 do include some more positive (and a few negative) evidence: 2013 (4%), 2014 (10%), 2015 (7%), 2016 (12%), 2017 (19%), 2018 (8%). The distribution over the years seems to be rather equal with a slight increase for the years 2016 and 2017 (Fig. 8). There

are only few negative results reported for this proposition. More studies (16%) explicitly emphasize the potential for LA to improve learning outcomes, but do not present any concrete improvements of learning outcomes (see e.g., Casquero, Ovelar, Romo, Benito, & Alberdi, 2013; Tabuenca, Kalz, Drachsler, & Specht, 2015; Tempelaar et al., 2017;

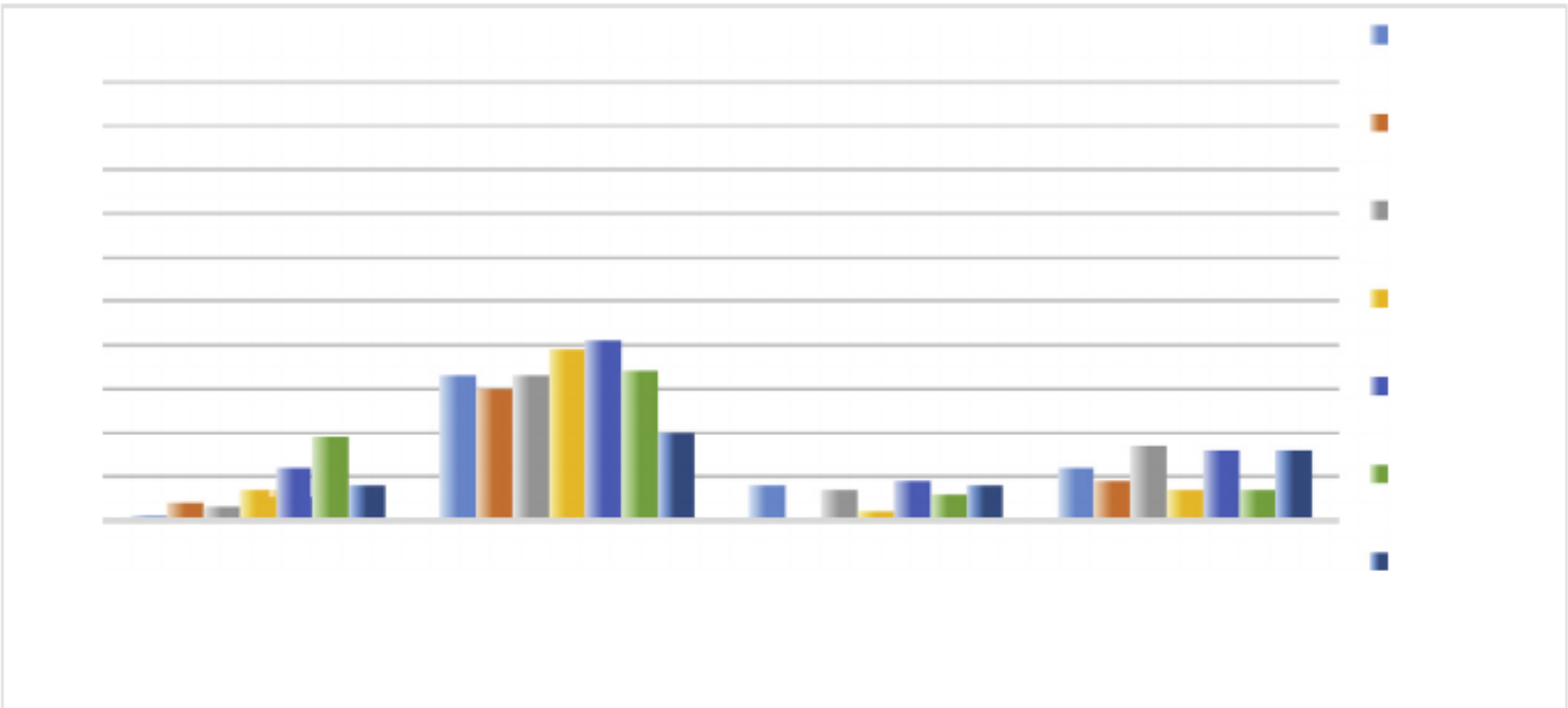


Fig. 7. Learning analytics evidence across the years 2012–2018 (%).

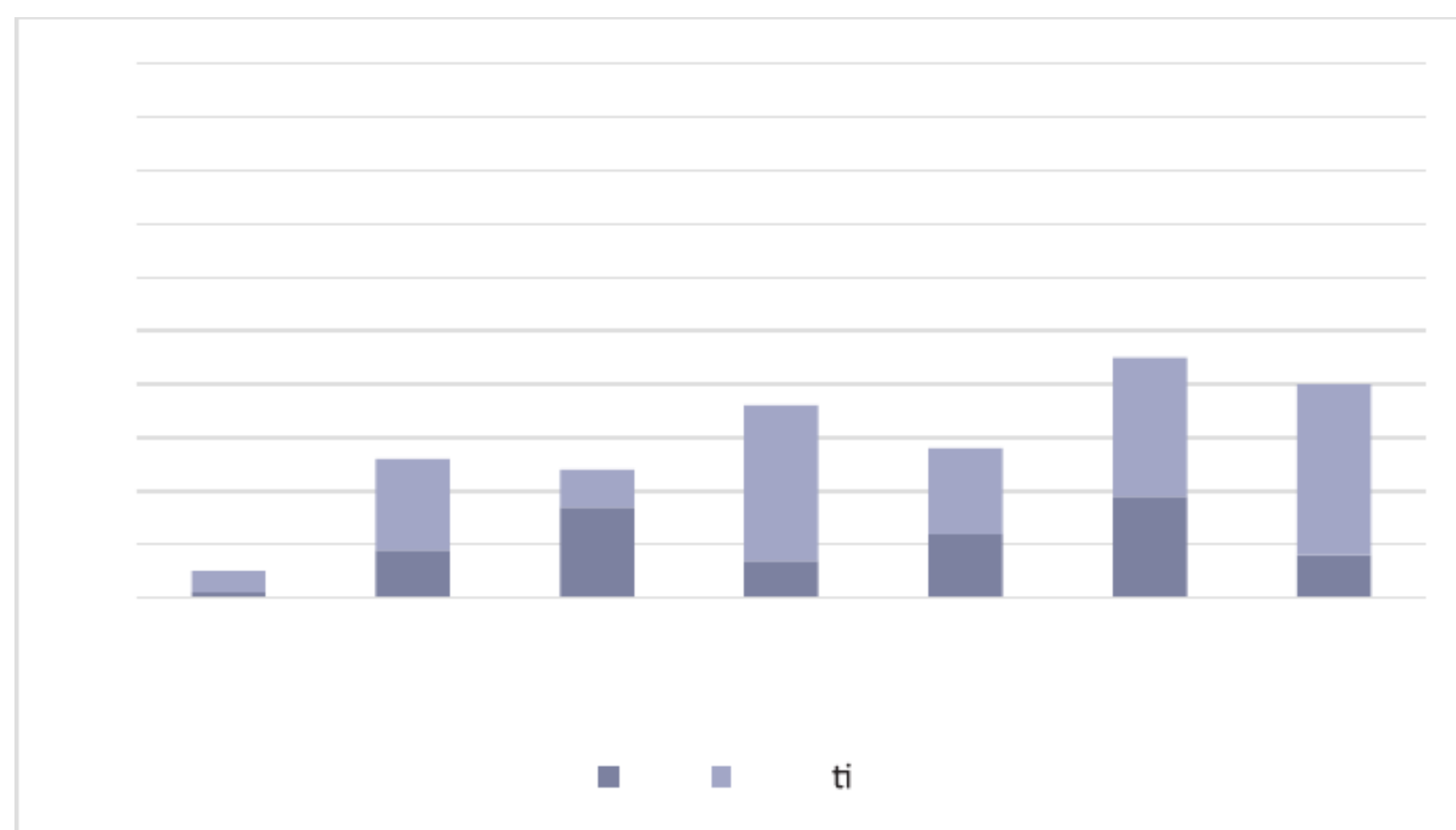


Fig. 8. Proposition 1. LA improve learning outcomes (%).

Wise, Perera, Hsiao, Speer, & Marbouti, 2012).

The studies' results that provide some evidence in improvements of learning outcomes focus mainly on three areas: i) *knowledge acquisition*, including improved assessment marks and better grades, ii) *skill development* and iii) *cognitive gains*.

3.4.1.1. Knowledge acquisition. Improvements in students' academic performance were identified by several researchers. Tempelaar, Cuypers, van de Vrie, Heck, and van der Kooij (2013), for example, explored the role of test-directed learning by investigating the intensity of use of two test-directed platforms and academic performance, which was measured through the exam containing a mathematics and statistics part and three quizzes. The results demonstrated that students benefited from the opportunity of test-directed learning. Others (Whitelock, Twiner, Richardson, Field, & Pulman, 2015) examined the use of a natural language analytics tool (OpenEssayist) to provide feedback to students when preparing an essay for summative assessment and showed that the students gained significantly higher overall grades than the students who did not have access to the tool. Moreover, Guarcello et al. (2017) by applying the Coarsened Exact Matching analytical approach to determine the impact of Supplementary Instruction (SI) participation on course performance - in particular the passing of the course - showed that higher exam performance by SI-attending students was attributable in part to the SI intervention. The intervention was designed to help students prepare for exams through active learning strategies and peer-facilitated study. In another example, Mangaroska, Sharma, Giannakos, Trættemberg, and Dillenbourg (2018), focused on understanding students' debugging behaviour using learner-centric analytics. The study's results showed that students who processed the information presented in the used tool that mirrored students' behaviour in programming tasks (focusing on debugging habits) and acted upon it, improved their performance and achieved higher level of success than those who failed to do it.

3.4.1.2. Skill development. Advances in students' time management skills, presentation skills, reflection and collaborative problem solving skills were exhibited. Improvements in students' time management skills were identified by Tabuenca et al. (2015), who investigated the effects of tracking and monitoring time devoted to learn with a mobile tool on self-regulated learning. Their findings showed positive effects and also revealed that notifications pushed at random time of the day do not produce significant improvements in time management. In a recent study, Ochoa et al. (2018) focused on the development of students' communication skills and presented a scalable system to

provide automatic feedback to entry-level students to develop basic oral presentation skills. The system's evaluation indicated that the system's feedback highly corresponded human feedback, and that students perceived the feedback useful to develop their oral presentation skills.

Furthermore, some LA studies have focused on problem solving skills. Worsley (2018), for example, by applying a qualitatively-informed multimodal LA approach presented some neutral evidence in the developments of the students' collaborative problem-solving skills. In particular, findings indicate that a) bimanual coordination (i.e., using both hands) seems to correlate with learning gains, b) participants who were able to realize the benefits of physical disengagement were only able to do so if they also completed a sufficient amount of being physically engaged, i.e., working with their hands. Kwong et al. (2017) focused the development of students' reflection skills, among others, through a game-based approach. By deploying learning trails, the "Trails of Integrity and Ethics" - immersed students in collaborative problem solving tasks centred on ethical dilemmas - students' interests in learning about issues of academic integrity and ethics in classrooms have been stimulated.

3.4.1.3. Cognitive gains. Evidence in terms of students' cognitive gains has been also found. For example, Gašević, Mirriahi and Dawson (2014) examined students' use of a video annotation tool. Two different instructional approaches were employed (i.e., graded and non-graded self-reflection annotations within two courses in performing arts) and exposed that students in the course with graded self-reflection adopted more linguistic and psychological related processes in comparison to the course with non-graded self-reflections. Chiu and Fujita (2014) investigated how statistical discourse analysis can be used to overcome shortcomings when analysing knowledge processes. The results showed how attributes at individual and message levels affected knowledge creation processes. Asynchronous messages created a micro-sequence context; opinions and asking about purpose preceded new information; anecdotes, opinions, different opinions, elaborating ideas, and asking about purpose or information preceded theorizing. These results present how informal thinking precedes formal thinking and how social metacognition affects knowledge creation. Finally, Sonnerberg and Bannert (2015) through the analysis of the effects of metacognitive prompts on learning processes and outcomes during a computer-based showed prompting increased the number of metacognitive activities, especially monitoring, which, in turn, increased the transfer performance.

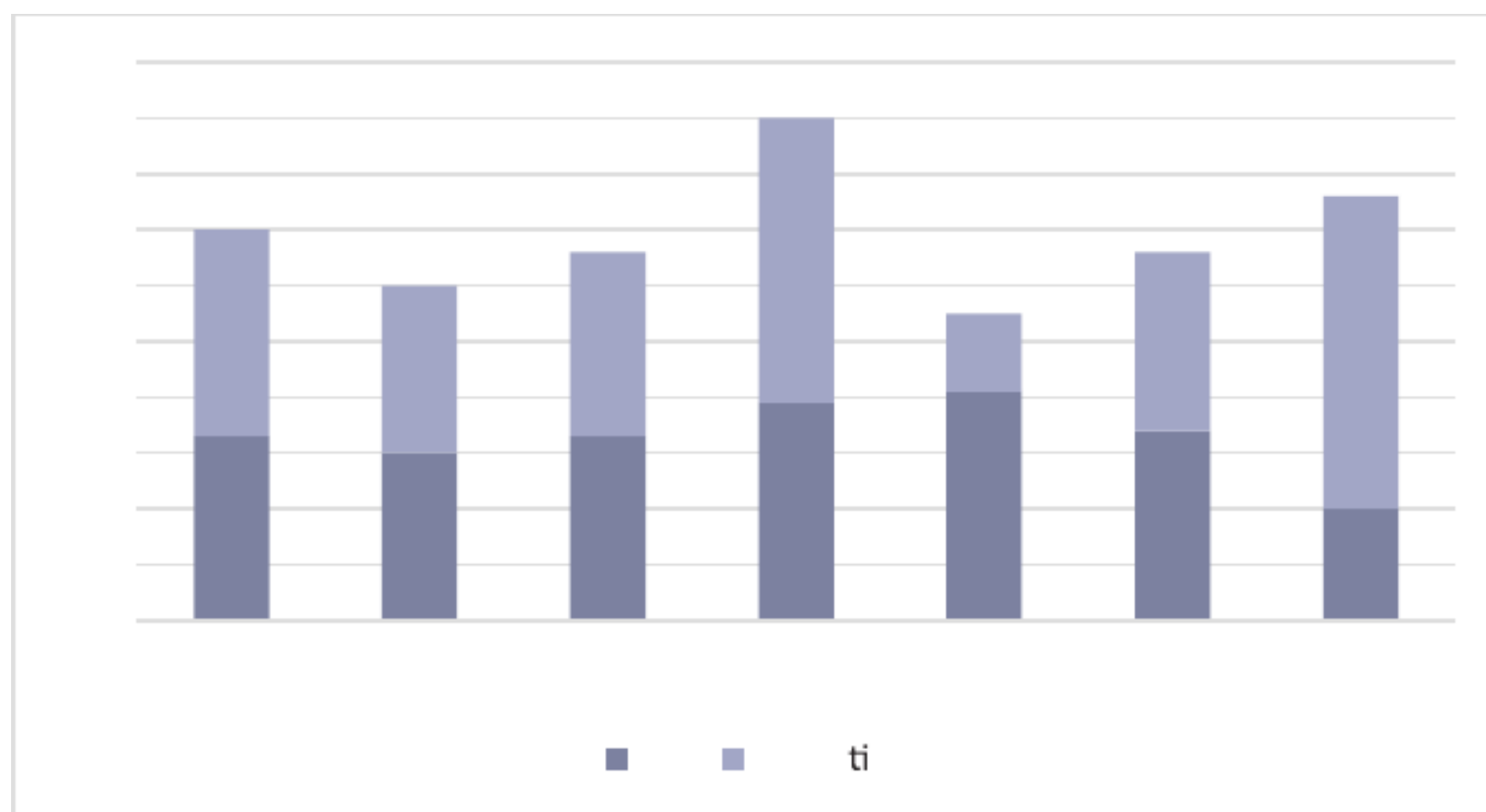


Fig. 9. Proposition 2. LA improve learning support and teaching (%).

3.4.2. Do LA improve learning support and teaching?

Of the four positions, *LA improve learning support and teaching*, including retention, progress and completion has the most evidence (35%; Fig. 6). The overall potential to improve learning support and teaching is even higher (62%), which suggests that we have to consider how to transfer this potential into practice. The distribution over the years 2012–2017 varies in general from 30% to 41% (Fig. 9).

Overall, our sample exhibits more evidence on supporting institutions and teachers, compared to the learners themselves.

In terms of supporting institutions, the Learning Analytics Readiness Instrument - aimed to prepare educational institutions for a successful analytics implementation – was introduced by Arnold, Lonn, and Pistilli (2014). Furthermore, its beta version was used to survey 560 participants representing 21 institutions (Oster, Lonn, Pistilli, & Brown, 2016) to understand the processes institutions use when discerning their readiness to implement LA. One of the latest studies (Tsai, Moreno-Marcos, Tammets, Kollom, & Gašević, 2018) presents the SHEILA (Supporting Higher Education to Integrate LA) framework informing institutional strategies and policy responses of LA.

LA researchers have also analysed student retention. While 13% of papers published between the years 2012 and 2016 had retention as their primary research focus, there is a decrease of such a focus for the last two years (see e.g., Casey, 2017; Chen, Johri & Rangwala, 2018). Overall, the findings show that the use of VLE data, assessment scores, institutional variables, differences in technology use, students' online engagement and course design and usage can predict, to various degrees, student retention. For example, course design with a heavy reliance on content and cognition (assimilative activities) seemed to lead to lower completion and pass rates (Rienties, Toetenel, & Bryan, 2015). Most predictive studies on student retention present and evaluate early warning systems (EWSs). Junco and Clem (2015), for example, showed that digital textbooks analytics are an effective EWS to identify students at risk of academic failure. Similarly, an EWS that supports students (Lonn et al., 2012) exposed whether mentors should take one of three actions: “encourage” students to keep doing well, “explore” students' progress in more detail, or immediately “engage” students to assess potential academic difficulties.

Learning design is another common research focus. In studies on learning design, most research was focused explicitly in terms of improving teacher support rather than improving learner support. Some researchers linked the learning design of courses with LMS usage and learning performance. Rienties et al. (2015), for example, found that learning design strongly influences how students engage online, where the students' LMS activity was considered as a proxy for student engagement. Another study examined how students used an adaptive

learning system based upon their needs and interests by analysing their use patterns (Liu et al., 2017). Students who had more prior knowledge (i. e., higher Diagnostic Test score) were found to make more attempts at testing themselves and had higher post-test scores. The results also indicated a lack of alignment between the assessments and the content students were directed to based upon their performance on the assessments. Several of the latest publications similarly focused various learning design aspects: e.g., the mismatch between learning design assumptions and students' actual behaviour (Nguyen et al., 2018), the visualisation of LA based on a learning design driven approach (Echeverria, Martinez-Maldonado, Granda, Chiluza, Conati & Buckingham Shum, 2018), and the use of worked examples (Tempelaar et al., 2017).

To improve teaching and learning support, LA research also focus on, and found some evidence for, understanding how students learn, using learner data to study behavioural practices such as learning styles, emotions, gestures and electro-dermal activation, speech, and online learner behaviour types. For example, multimodal learning analytics (MMLA) was applied to identify students' various behavioural practices in two different experimental conditions (principle-based vs. exam-based) in order to understand how learning interventions work (Worsley & Blikstein, 2015). The analyses revealed different behaviours that were related to principle-based reasoning, success and learning. Ezen-Can, Grafsgaard, Lester, and Boyer (2015) examined multimodal features related to posture and gesture in order to classify students' dialogue acts within tutorial dialogue for the purpose of improving natural language interaction. They found that these unsupervised models of students' dialogue acts' classification are significantly improved with the addition of automatically extracted posture and gesture information.

Finally, some evidence in terms of supporting development of learner models was also revealed. A new tool for visualisation of student learning model during a gameplay session was found to be effective (Minovic, Milavanovic, Sosevic, & Gonzales, 2015). The results of a statistical analysis indicated that the tool helped students in identifying and solving learning problems. Furthermore, the participants expressed that the tool gave good insight into individual student learning process as well as good foundation for tracking groups' progress. An alternative approach for dynamic student behaviour modelling grounded in the analysis of time-based student-generated trace data, with the purpose of classifying students according to their time-spent behaviour, was presented and tested by Papamitsiou, Karapistoli, and Economides (2016). The results indicate that a time-spent driven description of the students' behaviour could have an added value in dynamically reshaping the respective models.

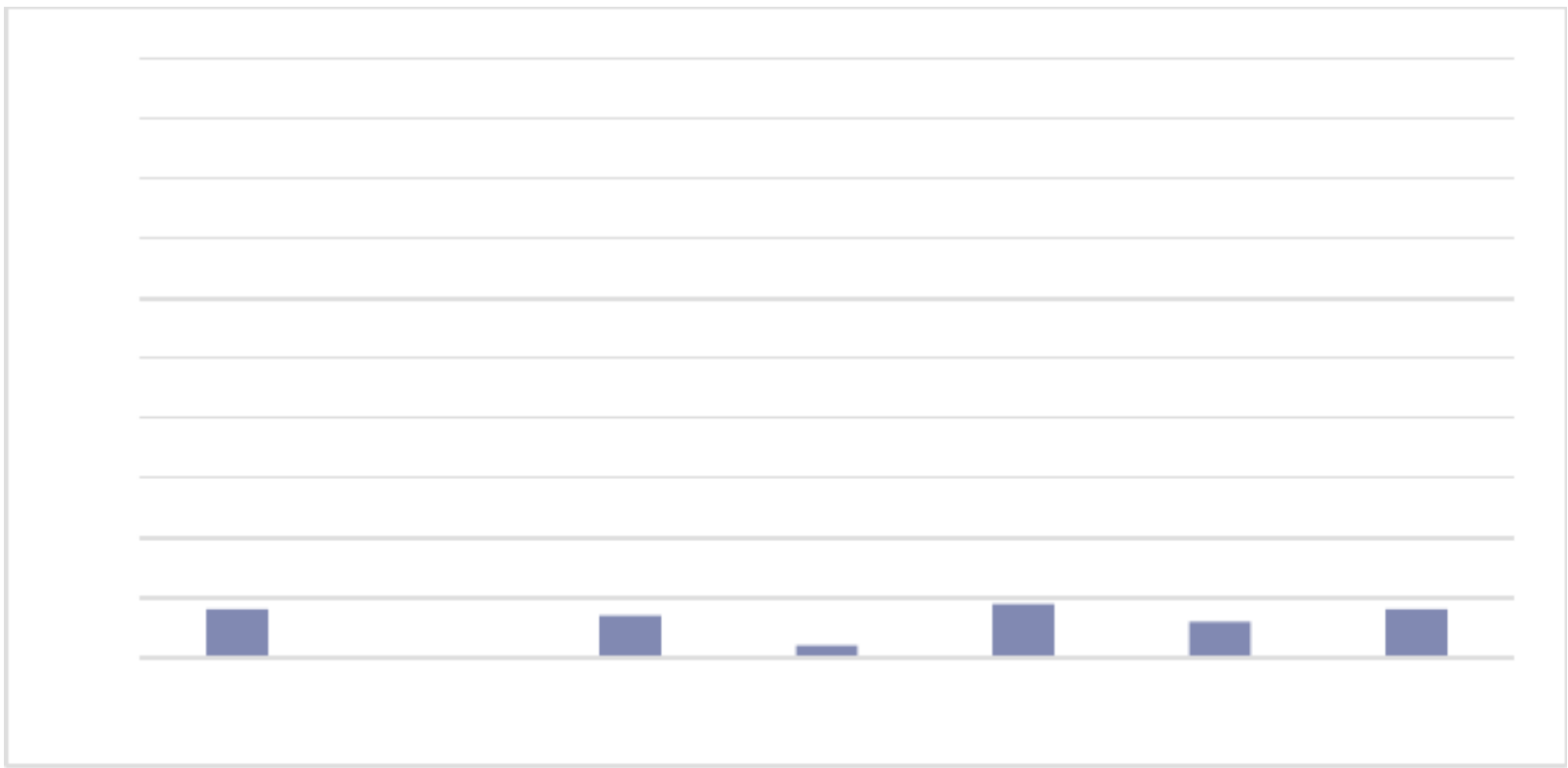


Fig. 10. Proposition 3. LA are used and taken up widely (%).

When considering the design of implementing LA, a Pedagogical Model for the LA Intervention, primarily aiming to support teachers (Wise et al., 2013) and the Align Design Framework (Wise, Vytasek, Hausknecht, & Zhao, 2016) were presented. The later was validated as a tool for learning design, supporting students’ use of analytics as part of the learning process.

3.4.3. Have LA be taken up and used widely?

This proposition focuses on the level of usage of LA, and is concerned with institutional and policy perspectives. Judging whether the proposed methods, systems, theories would scale to entire institutions or other institutions is not an easy task. We deemed the scalability condition as fulfilled only if is there is some deployment of the tool coupled with students or teachers’ involvement. According to this judgement, 94% of the papers do not scale (Fig. 6) and the distribution over the years does not indicate an increase over the years (Fig. 10). Rienties et al. (2015) presented one of the studies that was deployed at a large scale. They showed that learning design activities seem to have an impact on learning performance, in particular when modules rely on assimilative activities. Furthermore, Rienties and Toetenel (2016) exhibited that the development of relevant communication tasks that align with the course learning goals seems likely to enhance academic retention. The learning designs of four language education modules and online engagement of 2111 learners were contrasted using weekly learning design data by Nguyen, Rienties, and Toetenel (2017). The

findings from the study revealed that learning designs were able to explain up to 60% of the variability in student online activities. Herodotou et al. (2017) raised the need for devising appropriate intervention strategies to support students at risk. Kwong et al. (2017) reported the LA on the initial stages of a large-scale, government-funded project which inducts students in Hong Kong into consideration of academic integrity and ethics. A total of 658 undergraduate students from all disciplines from four institutions and 46 undergraduate student Hall Tutors participated in the study; the results indicated that situated learning using AR with mobile devices has been effective for students to learn abstract concepts of academic integrity and ethics. Moreover, Millecamp et al. (2018) introduced a learning dashboard that supports the dialogue between a student and a study advisor used at ten programs at one university. Another step up in scalability is taken by Broos et al. (2018) whose LA dashboard was run over several institutions, with consideration to e.g., data ownership. The dashboard was also used to provide feedback to students participating in a region-wide positioning test before entering a university program. In summary, there are very few studies that have been employed at scale over the years.

3.4.4. Have LA been used in an ethical way?

This proposition is about the ‘should we’ questions, rather than the ‘can we’ ones addressed by the other propositions. Overall only 18% of the articles mention or/and consider ethics and privacy in relation to

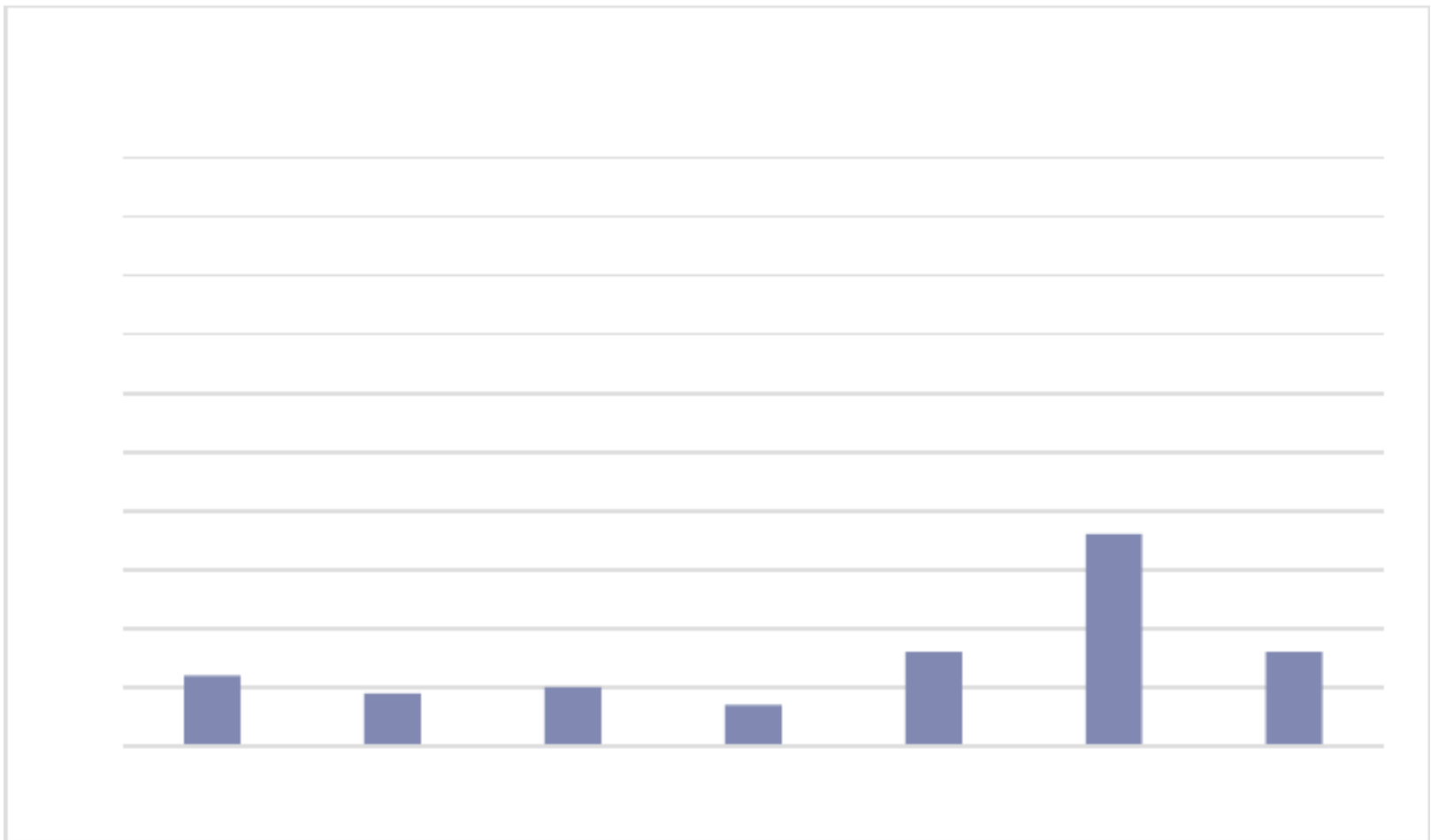


Fig. 11. Proposition 4. LA are used in an ethical way (%).

the conducted research, to a various degree. There is an increase for the year 2017–36% (Fig. 11) and there might be an increase for 2018: 16% of the LAK conference publications support evidence in this regard, compared to the previous years' LAK studies, where very few studies considered ethics.

Among the empirical studies where the issues of ethics have been a concern (e.g., data privacy and security, informed consent), only few papers approach them in a systematic way. Such examples include a study by Ferguson, Wei, He, and Buckingham Shum (2013), who considered ethics of visual literacy, assessment for learning and participatory design and deployment in their research. Another example presents the development of a LA dashboard by Broos et al. (2018), where the authors not only took into consideration data ownership, data privacy and portability issues, but these issues affected the technical design of the dashboard.

There are also a few papers that explicitly aimed to emphasize the issues of ethics and privacy. Slade and Prinsloo (2013), for instance, acknowledged ethical considerations in LA research and proposed six principles to guide higher educational institutions in addressing these issues. In another paper, Prinsloo and Slade (2016) explored students' vulnerability in relation to fostering the potential of LA. Rubel and Jones (2016) argued that LA poses moral and policy issues for students' privacy; there are crucial questions that need to be addressed to ensure that LA are commensurate with students' privacy and autonomy. Kwong et al. (2017) enlightened students on ethics by immersing them using augmented reality in collaborative problem solving tasks centred on ethical dilemmas, in real locations where such dilemmas might arise.

4. Discussion and conclusion

This study presented an extensive coverage of research on the use of LA in HE and examined this with a focus on the used research approaches and methods, as well the evidence of LA research for learning in HE.

Our analysis shows that the LA field is still an evolving area of practice and research in which descriptive studies and interpretative data collection methods prevail, in line with Papamitsiou and Economides (2014). However, the fact that 26% of the reviewed publications were categorised as 'theory use' studies, and 11% as 'theory generating' indicates that the field is maturing. We also found that only 19% of the reviewed studies used both qualitative and quantitative methods of data analysis, with an increase in 2017. This is a small number if we consider that good-quality research results - that would benefit learners and teachers in HE "for purposes of understanding and optimizing learning and the environments in which it occurs" (Long & Siemens, 2011, p. 34) - call for a much more frequent application of well-designed mixed-method approaches. A more frequent application of qualitative methods, in combination with quantitative ones, will help us to understand complex learning environments. This is similarly acknowledged by Ferguson and Clow (2017), who examined the evidence of LA in the subjects of psychology and education. The increase of mixed-methods for the year 2017 suggests that there might be a relevant development in this regard. Yet, to be able to claim that there is a stable trend, we need to investigate this further.

Predictive methods have been one of the dominating methods for several years. However, since 2016 the use of these methods has considerably decreased. The decrease, together with an increase of relationship mining methods and the rather stable use of methods for the distillation of data for human judgement, suggests that LA research in HE is shifting from prediction of, e.g., retention and grades, towards a deeper understanding of students' learning experiences.

Even though the above-mentioned shift is a positive sign, this study's results show that so far there is little evidence (9%) that the research findings demonstrate improvements in learning outcomes, including knowledge acquisition, skill development and cognitive gains, as well as learning support and teaching. Out of the four

propositions, *LA improve learning support and teaching*, has the most evidence, which confirms Ferguson and Clow's (2017) findings, based on a much smaller reviewed sample.

Despite the fact that the identified potential for improving learning support and teaching, as well as improvements in learning outcomes is high, we cannot currently see much transfer of the suggested potential into higher educational practice over the years, which would be expected. This raises a question of how we can facilitate this transfer process to benefit learners. In order to understand how LA's potential can be beneficial for HE and to provide guidelines for both researchers and practitioners, we need to examine this potential further, with a specific focus on opportunities, barriers and challenges.

Moreover, we need to rigorously consider what research questions we want to answer in order to improve learning practice in HE. Few have actually asked the question: what kind of data, that we may, or may not have today, would be valuable to analyse? In other words, we should carefully consider what data do we need and what we would like to do with the data. In this, the sciences of learning in combination with pedagogical knowledge might help.

Considering the limited number of studies that we deemed to be scalable (6%), it seems as the research has only used available data to see what is possible to conclude from a LA perspective. To be able to scale up LA research we have to consider several challenges, for example, data sharing, understanding complex learning environments, methods that would work efficiently and the generalisability of the presented results, i.e., to what extent can we rely on them in other educational settings? Will, for example, the identified predictors be the same in another course, another subject, another institution, another country? One way to progress in this path is to open up the used data sets for other researchers.

It is worrying that more than 80% of the papers do not mention ethics at all. Moreover, there are only few studies that approach ethical issues (e.g., data privacy and security, informed consent) in a systematic way. However, we should not jump to the conclusion that most studies are done in an unethical way, but we call for more explicit reflection on ethics to rise in the coming years. The increase of the studies that reflect on the ethical issues for the year 2017 (36%) might indicate that there is already a positive move in this direction.

This study has several limitations, which might be rather seen as the potential for future research. Firstly, we did not specifically focus on the analysis of the existing potential of the LA evidence. As the found potential is high, we need to examine it further in order to provide guidelines of how it could be transferred into the actual evidence. Secondly, our examination of the methods of data analysis focused mainly on computational methods. The further investigation of the other employed methods will contribute to a more holistic picture. Finally, we did not examine the actual use of theories. Thus, to better understand what theories have been used and in what ways in relation to the development of the LA research area and its impact on higher educational practice is a subject for future research. One exception is Jivet et al. (2018), who examined which theories from learning sciences have been integrated into the development of learning dashboards.

This study contributes to the literature by providing a systematic and detailed literature review of LA in HE. The results suggest that even though the LA field is maturing, the overall potential of LA is so far higher than the actual evidence, which poses a question of how we can facilitate the transfer of this potential into learning and teaching practice. Moreover, a further validation of developed LA tools, methods and more empirical cross-disciplinary and cross-institutional studies that are used in an ethical way are needed to deeper understand how LA can contribute to high quality and efficiency in HE. Our findings imply that the field is moving forward and we need to build on the existing literature instead of reinventing the wheel. We argue that this study's results would be a good starting point for researchers interested in the field of LA. For practitioners, the review act as an overview of what LA can contribute to academic institutions.

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